

What ReAct is, and how a tool call works

DATA 443/643 · Week 14, note 14.2

Week 14

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2026 L10 and Yao et al. Original slides; those sources are not copied.

Lecture plan (first time seeing ReAct)

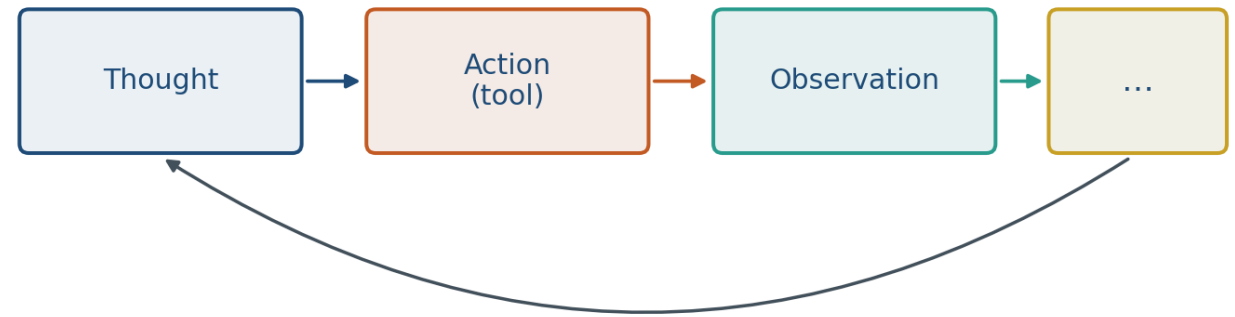
- 1 What it is, and why CoT is not enough
- 2 Architecture: policy, parser, tools
- 3 Formulas: sample, Tool(a), append
- 4 Pros, cons, unfaithful next-thought
- 5 Worked calc[17*24] → 408

Key goal: define ReAct, write the five-line log, and say why the observation must be in the next context.

What ReAct is

- Thought: scratch in language. Action: a structured tool call. Observation: a string the model did not write.
- CoT can invent 17×24 . ReAct is supposed to wait for calc.
- RAG retrieve is one tool. calc is another. Same loop.

ReAct: think, call a tool, read the result, repeat



Why a tool, not more thought tokens

Chain of thought

Extra tokens before the answer. Still guesses. Fluency is not a check.

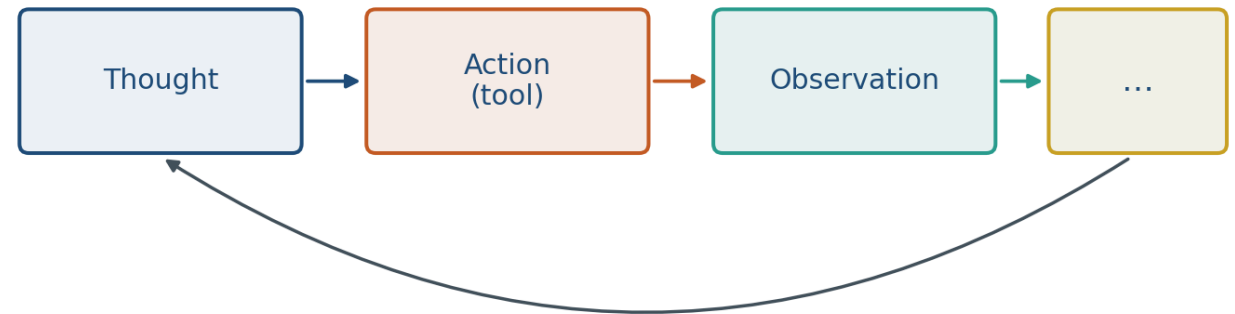
ReAct

The next thought can depend on a real observation: arithmetic, search, or a retrieved chunk.

Architecture

- Policy: the LLM samples the next line.
- Parser: action → tool name and arguments. Failures return ERR.
- Tools: functions. Lab 14 calc only matches digits and +*./.

ReAct: think, call a tool, read the result, repeat



The loop is a decoding protocol

$$o_t = \text{Tool}(a_t), \quad c_{t+1} = c_t \oplus \tau_t \oplus a_t \oplus o_t$$

$(\tau_t, a_t) \sim p_\theta(\cdot \mid c_t)$. No
new trained module today.

Stop at Answer, or at a step cap.

Stuffed RAG is one retrieve then one
generate. ReAct can call again.

What it is good and bad at

Positive

Delegate exact ops. Mix retrieve and calc. The transcript is an audit log.

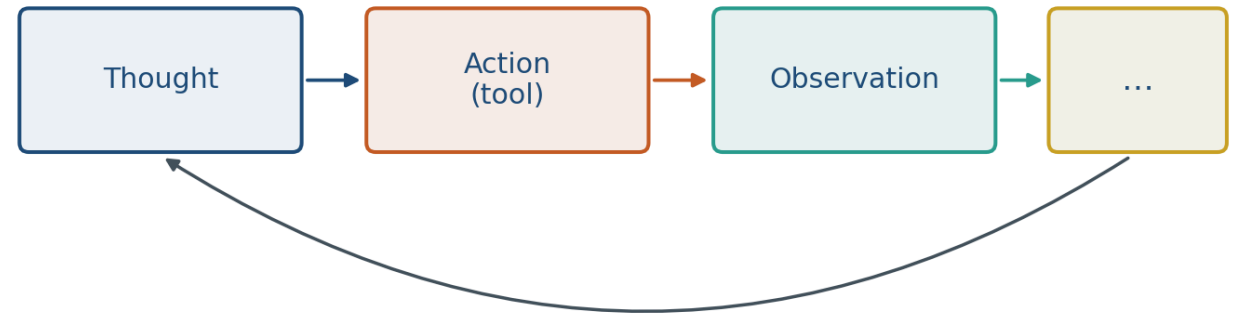
Negative

Bad parses, wrong tool, ignored obs, loops, latency. Free Python eval is out of bounds in this class.

Do not multiply in the thought

- Action `calc[17*24]`. Observation 408.
Answer 408.
- If the next thought says 428, the loop is unfaithful.
- Lab 14: `calc[23*60]` → 1380, plus a citation [d1].

ReAct: think, call a tool, read the result, repeat



Three control loops

CoT / SC / ToT

Text only, or search among thoughts. No observation from the world unless you add one.

ReAct

Interleave thoughts with tool results. Missing a line in the log means you ran CoT.

What to take from this lecture

Definition

Think, call a tool, read, continue.

Five-line log

Thought, Action, Observation, Thought, Answer.

Dummy calc

Lab 14 is not `python_eval`.

Cap

Looping `calc[1+1]` is a missing step limit.